Subject: Electronics 1
Duration: 50 min

Mid Exam

Spring 2025/2026

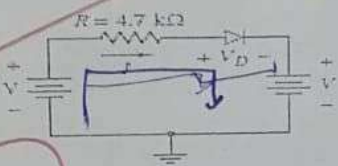
Q1: (a) In the circuit shown in the figure, find the diode voltage V_D and the supply voltage V such that the current is $I_D = 0.4mA$. Assume the diode threshold voltage is $V_T = 0.7V$. (b) Using the results of part (a), determine the power dissipated in the diode. (6 marks)

in KVL

in forword $V_D = V_T = 0.7$.

if I not zero $V_D = 0.7$ not o.c

cs; $V_D = 0.7V$



$-5V + (4.7k)(0.4mA) + 0.7 + V = 0$

$\boxed{V = 2.42V}$ (a)

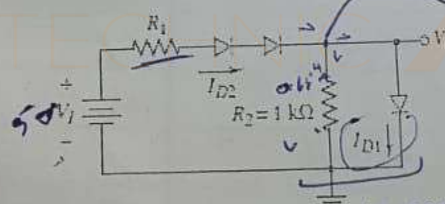
$I_D = I_R = I$ in series

+6

$P_D = I_D \times V_D$
$= 0.4mA \times 0.7V$
$= 0.28\ mW$

Q2: Assume each diode in the circuit shown in Figure has a threshold voltage of $V = 0.65V$. (a) The input voltage is $V_I = 5V$. Determine the value of R_1 required such that I_{D1} is one-half the value of I_{D2}. What are the values of I_{D1} and I_{D2}? (6 marks)

$I_{D1} = \frac{1}{2} I_{D2}$



Iin = Isum

in parallel

$V_R = V_D = 0.65$ $I_{R_2} = \frac{0.65}{1k}$
$= 0.65\ mA$

R?

+4

$I_{D2} = 0.65mA + I_{D1}$

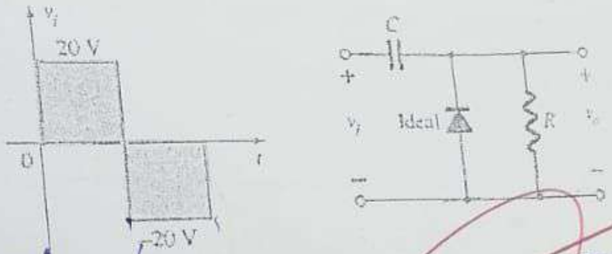
$I_{D2} = 2(0.65mA)$ $I_{D2} = 2I_{D1}$

$I_{D1} = 1.3\ mA$

$2I_{D1} = 0.65mA + I_{D1}$

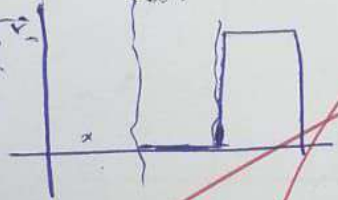
$\boxed{I_{D1} = 0.65\ mA}$

Q3: Sketch V_o for the network of Fig. for the input shown. (6 marks)



$V_{p-p} = 40$

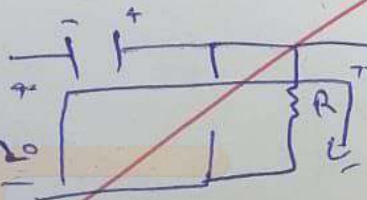
$20 + 20 = 40$



$V_{p-p} = 40\ V$

$20 \to V_C$

+ve

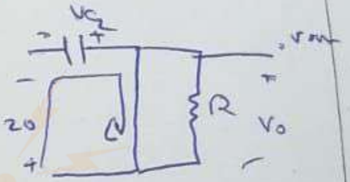


$V_o = 40\ V$

$-20 - 20 + V_o = 0$

+6

→ +ve



$20\ V - V_C = 0$

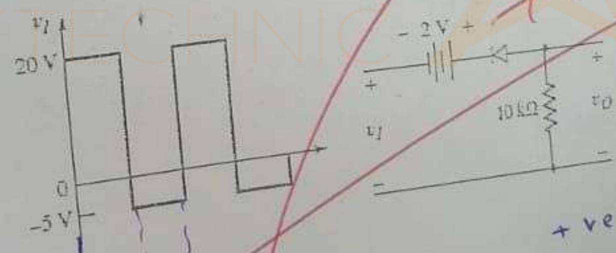
$V_C = 20\ V$

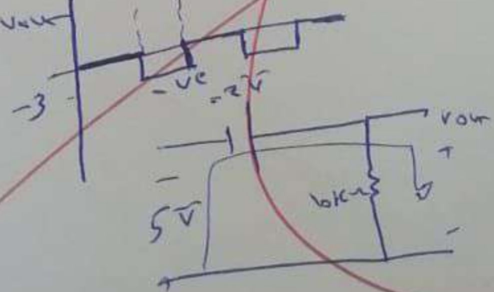
$IR = V_R = V_{out} = 0$

$I = 0$

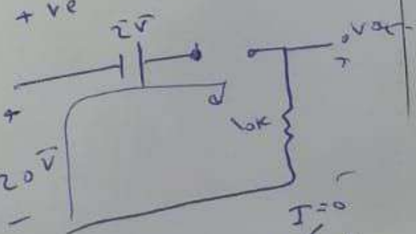
Q4: Sketch v_0 for the circuit in Figure (6 marks)

+6





+ ve



$I = 0$

$V_o = V/R = 0$

$-20 - 2 + V_{oc} = 0$

$V_{oc} = 22$

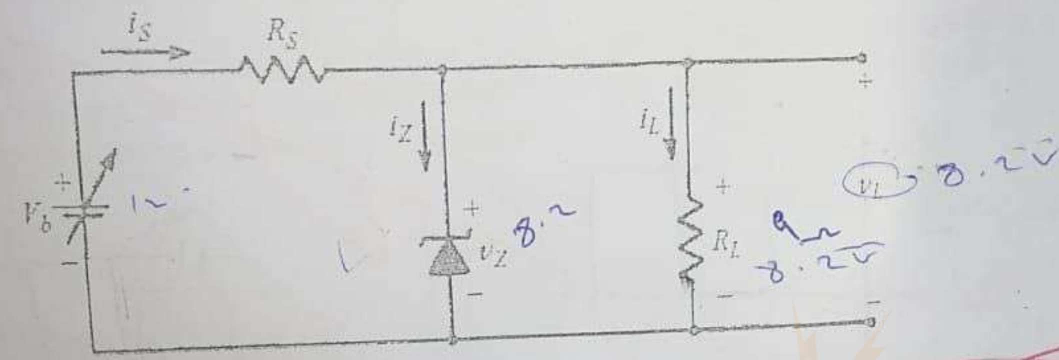
$V_o = 0$

$5\ V - 2\ V + V_o = 0$

$V_o = -3\ V$

Q5: The Zener diode in the voltage-regulator circuit of the following Fig. has a constant reverse breakdown $V_Z = 8.2V$, for $75mA \le i_Z \le 1A$. If $R_L = 9\Omega$, find R_S so that $v_L = V_Z$ is regulated to 8.2V while V_b varies by +10 percent from its nominal value of 12 V. (6 marks)

$I_{Z min}$ $I_{Z max}$



$R_1 = 3.85\Omega$
when $V_b = 12$

$I_{Lmin} = I_s - I_{Zmax}$

$I_L = \frac{V_L}{R}$

$I_L = \frac{8.2}{9} = 0.911 A$

$I_{Zmin} = 75 mA$

$I = \frac{V_{in} - V_Z}{R_s}$

$I_{Lmax} + I_{Zmin} = I_S$

$= 75mA + 911 mA$

$9.82 = \frac{12 - 8.2}{R_s}$

$I_s = 986 mA$

$R_s = \frac{1.8}{986 mA}$

when $R_s = 1.8255 \Omega$

$V_b = 10 V$

$-10 + V_s + 8.2 = 0$

$V_s = 1.8$

$I_s R_s = 1.8$

$I_s - 986 mA$

$= 1.825 \Omega$